Core Java Practice Programs

55 programs arranged from Level 1 to Level 5, from easy to hard.

Use this list to revise syntax, loops, arrays, strings, OOP, exception handling, collections, file handling, multithreading, and JDBC.

Level 1 - Very Easy Basics

1. Print your name, age, and city
2. Swap two numbers using a third variable
3. Swap two numbers without using a third variable
4. Check whether a number is even or odd
5. Find the largest of two numbers
6. Find the largest of three numbers
7. Print numbers from 1 to 100
8. Print even numbers from 1 to 100
9. Print odd numbers from 1 to 100
10. Calculate the sum of first N natural numbers
11. Print multiplication table of a number

Level 2 - Easy to Moderate

12. Check whether a number is prime
13. Print all prime numbers between 1 and N
14. Check whether a number is palindrome
15. Check whether a number is Armstrong number
16. Reverse a number
17. Count digits in a number
18. Find factorial of a number
19. Find Fibonacci series up to N terms
20. Find the sum of digits of a number
21. Check whether a number is perfect
22. Print pattern of stars in pyramid or triangle form

Level 3 - Arrays and Strings

23. Find the largest and smallest element in an array
24. Calculate sum and average of array elements
25. Reverse an array
26. Search an element in an array
27. Sort an array in ascending order
28. Find second largest element in an array
29. Remove duplicates from an array
30. Merge two arrays
31. Count vowels, consonants, digits, and spaces in a string
32. Reverse a string

33. Check whether a string is palindrome
34. Find frequency of each character in a string

Level 4 - OOP and Core Java Concepts

35. Create a Student class with data members and methods
36. Create a constructor and use constructor overloading
37. Use this keyword in a class
38. Use static variable, method, and block
39. Implement inheritance with Person and Employee classes
40. Demonstrate method overloading
41. Demonstrate method overriding
42. Create an abstract class and extend it
43. Create an interface and implement it
44. Use final keyword in variable, method, and class
45. Create a package and import classes from it

Level 5 - Hard and Advanced Core Java

46. Handle arithmetic exception using try-catch
47. Create custom exception
48. Use finally, throw, and throws
49. Read and write data using FileReader and FileWriter
50. Copy content from one file to another
51. Count words, lines, and characters in a file
52. Use ArrayList and perform add, remove, search, sort
53. Use HashMap to store and display student marks
54. Create two threads using Thread class and Runnable interface
55. Build a simple JDBC program to connect Java with MySQL and fetch records

Suggested order: finish Level 1 and Level 2 first, then move to arrays/strings, OOP, and finally advanced Core Java topics.